

Exploratory Data Analysis

OPTN Kidney Waitlist Outcomes

Predicting Organ Transplant Waitlist Outcomes & Optimizing Allocation

Organization	MediMate Foundation
Program	Break Through Tech AI Studio - Fall 2026
Dataset	OPTN Kidney Waitlist Analytic Extract
Analysis Window	January 1, 2015 through June 30, 2026
Report Purpose	Data understanding, quality assessment, modeling readiness, fairness and survival-analysis planning

Prepared from the completed exploratory analysis of 494,862 kidney waitlist registrations.

Executive Summary

This exploratory data analysis examined the OPTN kidney waitlist analytic dataset to understand the population, outcomes, data quality, missingness, temporal patterns, geographic differences, and variables that may be useful for predicting adverse waitlist outcomes and time to transplant.

The working dataset contains 494,862 kidney waitlist registrations representing 414,187 unique patients across 256 listing centers. One row represents one waitlist registration, not necessarily one patient, so some patients appear more than once.

The primary classification target is Adverse Event ('event_adverse'), defined as death while waiting or removal because the patient became too sick for transplant. There are 69,743 adverse registrations, or 14.09% of the dataset. The remaining 85.91% do not have one of these two recorded adverse outcomes.

Bottom line

The dataset is large and suitable for classification, survival analysis, explainability, and fairness work, but successful modeling depends on handling censoring, repeated patients, temporal missingness, post-outcome leakage, invalid BMI values, and geographic/center heterogeneity correctly.

Most important findings

1. Age and functional status show strong relationships with adverse outcomes. Older candidates and candidates with poorer functional status have substantially higher observed adverse-event rates.
2. Dialysis status is important. Patients already on dialysis when listed have a 15.51% adverse-event rate compared with 10.28% among those not yet on dialysis.
3. Geographic variation is substantial. Both adverse-event and transplant rates differ across OPTN regions and even more strongly across individual listing centers.
4. Blood type is strongly related to observed transplant patterns. Blood Types O and B have lower transplant rates and longer median waits than A and AB.
5. Censoring is a major issue. Recent candidates have much shorter follow-up and are much more likely to still be waiting, so raw outcome percentages cannot be compared across years without accounting for observation time.
6. Initial CPRA ('INIT_CPRA') has a major temporal missingness problem: it is fully recorded from 2015 through 2022 but roughly 80% missing from 2023 onward.
7. Body Mass Index ('BMI_TCR') contains clear invalid extreme values, including values above 400,000.
8. The same patient may appear in multiple registrations. Future train/test splitting must therefore be grouped by Patient ID ('PT_CODE') to avoid patient leakage.
9. Several variables occur only after transplantation or at the end of the waitlist episode. These can support descriptive analysis but must not be used as listing-time predictors.

1. Dataset Structure and Unit of Analysis

Item	Result
Waitlist registrations	494,862
Unique patients	414,187
Unique waitlist registration IDs	494,862
Variables	38
Listing centers	256
Earliest listing date	January 1, 2015
Latest listing date	June 30, 2026
Duplicate Waitlist IDs ('WL_ID_CODE')	0

The unit of analysis is one waitlist registration. Patient ID (`PT\_CODE`) identifies a person, while Waitlist Registration ID (`WL\_ID\_CODE`) identifies a specific listing episode.

Repeated registrations

Registrations per patient	Number of patients
1	349,609
2	52,358
3	9,463
4	2,013
5	495
6	168
7	53
8	17
9	6
10	3
11	2

Modeling implication

Randomly splitting rows could place different registrations from the same patient in both training and test sets. Classification and survival models should use a patient-grouped split based on Patient ID (`PT\_CODE`).

Documentation reconciliation

The repository documentation and the current compressed analytic file are slightly out of sync. The extraction summary reports 40 columns, while the actual dataframe contains 38. The documented `WL\_ORG` and `WLKI` columns are absent from the current CSV. Because the cohort is already restricted to kidney-alone registrations and `WLKI` was documented as fully missing, this does not materially affect the current analysis. The documentation also refers to 11 listing-time features while the current dictionary explicitly lists 10. The current CSV should be treated as the operational source of truth while these discrepancies are recorded for the team.

2. Outcomes, Targets, and Class Balance

Outcome	Count	Percent
Transplanted	234,429	47.37%
Still waiting	103,300	20.87%
Removed administratively	45,332	9.16%
Transplanted elsewhere	37,243	7.53%
Removed too sick	35,219	7.12%
Died	34,524	6.98%
Unknown	4,815	0.97%

The primary adverse classification target is Adverse Event (`event\_adverse = 1`) = Died + Removed Too Sick.

Classification target	Count	Percent
No recorded adverse event	425,119	85.91%
Died / removed too sick	69,743	14.09%

The derived target fields are internally consistent: died and removed-too-sick records map to `event\_adverse = 1`, transplanted records map to `event\_transplant = 1`, and still-waiting records map to `censored = 1`. However,

`event\_adverse = 0` is heterogeneous and includes patients who are still waiting, administratively removed, transplanted elsewhere, or in the unknown group.

Metric implication

Because only 14.09% of registrations are positive for the adverse target, accuracy alone would be misleading. PR-AUC, ROC-AUC, Brier score, calibration, sensitivity/recall, precision, and top-decile lift should be emphasized.

3. Censoring, Follow-Up, and Time Trends

Censoring is one of the defining characteristics of this dataset. A candidate marked Still Waiting has not permanently avoided an adverse event; the observation window ended before a final event occurred.

Listing year	Still waiting	Median observed days
2015	0.75%	737
2016	1.17%	700
2017	1.77%	654
2018	2.60%	629
2019	4.07%	610
2020	6.57%	563
2021	10.80%	538
2022	16.90%	506
2023	25.79%	470
2024	38.66%	474
2025	59.29%	269
2026	83.34%	74

The raw adverse-event rate appears to fall from 22.74% in 2015 to 0.67% in 2026. This should not be interpreted as evidence that recent patients became dramatically safer. Recent cohorts simply have much less follow-up time. Even when only resolved records are examined, recent years retain lower adverse rates, so shorter follow-up and changing patient/policy conditions remain important.

Survival-analysis implication

Survival methods are essential because they explicitly account for incomplete follow-up. For classification, treating every currently waiting patient as a permanent negative could create label bias. A fixed prediction horizon or follow-up eligibility rule should be considered.

Listing volume over time

Year	Registrations
2015	36,599
2016	36,956
2017	37,209
2018	40,564
2019	43,223
2020	37,825
2021	42,130
2022	44,442
2023	47,003
2024	49,635
2025	53,231

Year	Registrations
2026	26,045 (partial year)

Registrations generally increased over time, with a noticeable 2020 dip. The 2026 total is partial because the dataset ends June 30, 2026.

4. Listing-Time Features and Missingness

Plain-English feature	Column	Missing	Missing %
Age at Listing	`INIT_AGE`	0	0.00%
Sex	`GENDER`	0	0.00%
Race/Ethnicity	`ETHCAT`	0	0.00%
Blood Type	`ABO`	0	0.00%
Initial CPRA	`INIT_CPRA`	142,741	28.84%
On Dialysis at Listing	`ON_DIALYSIS`	0	0.00%
Body Mass Index	`BMI_TCR`	1,558	0.31%
Functional Status	`FUNC_STAT_TCR`	3,975	0.80%
Initial Medical Status	`INIT_STAT`	0	0.00%
OPTN Region	`REGION`	0	0.00%

A total of 350,337 registrations, or 70.79%, contain all 10 listing-time features. Missing Initial CPRA is by far the main reason records are incomplete. Body Mass Index and Functional Status have low missingness, and only 234 registrations are missing both.

5. Initial CPRA: Strong Temporal Missingness Shift

Initial CPRA (`INIT\_CPRA`) is a 0-100 sensitization measure that roughly reflects how difficult it may be to find a compatible donor. Lower values generally mean compatibility with more potential donors; higher values mean compatibility with fewer.

Listing year	Initial CPRA missing
2015-2022	0%
2023	79.53%
2024	82.86%
2025	81.15%
2026	80.76%

This missingness is not random. Something changed in recording, extraction, or field availability around 2023. Blind median imputation across the full dataset would hide a real structural shift.

- Investigate OPTN field availability or extraction changes beginning in 2023.
- Consider an Initial CPRA missingness indicator.
- Include listing year or policy era in sensitivity analyses.
- Compare models with and without Initial CPRA.
- Do not treat missing post-2022 CPRA as CPRA = 0.

CPRA change during the waitlist episode

Among 303,297 registrations with both Initial CPRA (`INIT\_CPRA`) and End CPRA (`END\_CPRA`): CPRA increased in 125,365, stayed the same in 162,466, and decreased in 15,466. The median change was 0 and the mean change was +15.6 points. End CPRA is measured later and must not be used as a listing-time predictor.

When analysis was limited to 2015-2022, where Initial CPRA was completely recorded, adverse rates were fairly similar across most CPRA groups. The highest CPRA group (98.01-100) had an adverse rate of 21.11%, but the overall EDA did not show a simple monotonic relationship between CPRA and the adverse classification target.

6. Patient Characteristics and Adverse Outcomes

Age at Listing (`INIT\_AGE`)

Age ranges from 0 to 91 years. Pediatric candidates are genuinely present, so very low ages should not automatically be treated as data errors. Mean age is approximately 51.9 years and median age is 54 years.

Age group	Adverse rate
0-17	2.00%
18-29	4.25%
30-39	6.69%
40-49	10.72%
50-59	15.11%
60-69	19.85%
70-79	20.43%
80+	18.79%

Key finding

Age at Listing (`INIT\_AGE`) is one of the strongest univariate relationships observed with the adverse outcome. The 80+ group is small (729 registrations), so its slight decline should be interpreted cautiously.

Sex (`GENDER`)

The dataset contains 306,408 male and 188,454 female registrations. The observed adverse-event rate is 14.74% for males and 13.05% for females. This is an association, not a causal conclusion, and may reflect differences in age, diagnosis, dialysis exposure, geography, or other factors.

Race/Ethnicity (`ETHCAT`)

Group	Adverse rate
American Indian	17.02%
Pacific Islander	14.55%
White	14.51%
Black	14.22%
Hispanic	13.49%
Asian	13.22%
Multiracial	12.55%
Unknown	3.99%

The Unknown group is small and behaves differently, so it should not be interpreted as a meaningful demographic pattern. Raw subgroup outcome differences are not evidence of model unfairness by themselves; fairness should be assessed using subgroup discrimination, calibration, sensitivity, and error rates.

Dialysis Status (`ON\_DIALYSIS`)

At listing, 360,617 registrations are already on dialysis and 134,245 are not. The adverse-event rate is 15.51% among those on dialysis versus 10.28% among those not on dialysis - a difference of more than five percentage points.

Key finding

Dialysis status appears to be an important listing-time predictor of adverse outcomes.

7. Dialysis Date and Allocation Priority Time

Dialysis Start Date (`DIALYSIS\_DATE`) has meaningful structured missingness: every registration with On Dialysis = No lacks a dialysis date, while every registration with On Dialysis = Yes has one.

A total of 318,009 registrations had dialysis start dates before the official Listing Date (`INIT\_DATE`). Among these patients, median pre-listing dialysis time was 561 days and mean time was approximately 906 days.

This explains why Allocation Priority Days (`DAYSWAIT\_ALLOC`) are often greater than Total Days on Waitlist (`DAYSWAIT\_CHRON`). For 317,966 comparable registrations, 295,750 had an exact match between extra allocation credit and pre-listing dialysis time, 296,169 matched within 30 days, and 15,086 differed by more than one year.

Interpretation

For most candidates, extra allocation waiting-time credit is essentially explained by eligible dialysis time before formal listing. The remaining large differences should be investigated as possible policy-credit, historical-rule, or data-recording effects rather than automatically labeled as errors.

8. Blood Type and Transplant Patterns

Blood Type (`ABO`) is strongly related to observed transplant patterns.

Blood type	Observed transplant rate	Median wait among transplanted
AB	62.61%	170 days
A	53.39%	221 days
B	44.24%	285 days
O	43.16%	276 days

Observed adverse rates were 15.01% for O, 14.85% for B, 12.82% for A, and 10.06% for AB. Rare A subtypes (A1, A2, A1B, A2B) have much smaller sample sizes and should be interpreted cautiously.

Key finding

Blood Types B and O have lower observed transplant rates and longer median waits among transplanted candidates than A and AB. Blood type is therefore important for the transplant-time component of the project.

A2/A2B-to-B Eligibility (`A2A2B\_ELIGIBILITY`)

This field is 92.27% missing overall, but the missingness is structured rather than general data loss. It is recorded only for Blood Type B candidates: 38,258 B candidates have eligibility recorded and 35,778 B candidates are missing it; all other blood types are missing by design/context.

9. Functional and Medical Status

Functional Status (`FUNC\_STAT\_TCR`)

Functional Status shows one of the strongest relationships with adverse outcomes. On the main adult scale, higher functional codes generally correspond to better functioning and lower observed adverse rates.

Functional code	Adverse rate
2010	27.76%
2020	24.15%
2030	23.25%
2040	18.34%
2050	16.98%
2060	18.00%
2070	16.38%
2080	13.58%
2090	12.32%
2100	10.94%

The alternate 4010-4100 scale also broadly shows decreasing adverse rates at higher functional levels, but those groups are much smaller. Code 998 represents unknown. One record contains `FUNC\_STAT\_TCR = 1`, which does not fit the expected coding scheme and should be investigated or treated as invalid if the source dictionary does not support it.

Key finding

Better functional status at listing is strongly associated with lower observed adverse-event risk.

Initial Medical Status (`INIT\_STAT`)

Initial Medical Status is heavily concentrated in codes 4010 (348,947 registrations) and 4099 (145,670 registrations). Their adverse rates are 14.31% and 13.56%, respectively. Remaining codes have very small sample sizes. In simple univariate analysis, Initial Medical Status appears less informative than age, dialysis status, or functional status.

10. Body Mass Index: Data-Quality Issue and Nonlinear Pattern

Body Mass Index (`BMI\_TCR`) has a plausible median of 28.53, but the raw distribution contains clearly impossible extremes. The maximum is approximately 430,226.67.

Quality flag	Count
BMI <= 10	240
BMI > 60	474
BMI > 100	263
BMI > 1,000	124

The very low BMI values are also suspicious because most affected records are adults; the median age among BMI <= 10 records is approximately 54 years. This indicates data-quality problems rather than a pediatric explanation.

BMI group (descriptive valid-range check)	Adverse rate
10-18.5	7.88%
18.5-25	12.66%
25-30	14.42%
30-35	15.41%
35-40	14.70%
40-50	13.18%
50-60	15.73%

Cleaning implication

Do not allow impossible BMI values to distort model training. A clinically defensible valid range should be agreed with the advisor/domain experts, and clearly invalid values should likely be flagged and treated as missing rather than blindly winsorized.

11. Waitlist Duration and Date Validation

Total Days on Waitlist (`DAYSWAIT\_CHRON`) and Derived Time to Outcome (`days\_to\_event`) agree extremely well. Among records where both are available, 494,798 are exactly equal and only 5 have different numeric values.

Observed duration	Adverse rate
0-3 months	5.26%
3-6 months	7.49%
6-12 months	10.98%
1-2 years	16.05%
2-3 years	19.72%
3-5 years	21.95%
5+ years	22.35%

This should not be read causally: longer observation creates more opportunity for adverse outcomes and is intertwined with censoring and selection. `days\_to\_event` and `DAYSWAIT\_CHRON` are post-listing duration variables and must not be used as baseline classification predictors.

Date errors

There are 59 registrations where End Date (`END\_DATE`) occurs before Listing Date (`INIT\_DATE`). Of these, 58 are labeled died and 1 is an administrative removal. The pipeline already sets `days\_to\_event` to missing for these records. They should be excluded from survival-duration modeling unless the dates can be corrected from the source.

Observed time by outcome

Outcome	Median days	Mean days
Transplanted	249	480.84
Transplanted elsewhere	503	675.50
Still waiting	520	743.78
Died	697	844.82
Unknown	700	890.98
Removed administratively	736	881.30
Removed too sick	805	946.22

Candidates removed for becoming too sick generally have the longest observed durations, while successful transplants occur earlier. This is descriptive context, not a causal statement.

12. Geographic Variation by OPTN Region

Geography is one of the strongest system-level findings in the EDA.

Regional metric	Highest	Lowest
Adverse-event rate	Region 1: 16.95%	Region 7: 11.58%
Transplant rate	Region 8: 55.62%	Region 4: 42.47%

Regional metric	Highest	Lowest
Still-waiting rate	Region 5: 26.79%	Region 10: 16.39%
Median wait among transplanted	Region 1: 400.5 days	Region 4: 211 days

The adverse-rate spread is about 5.4 percentage points, while the transplant-rate spread exceeds 13 percentage points. Importantly, a region can have a low overall transplant rate yet a short waiting time among patients who do get transplanted. Region 4 is an example: its transplant rate is low, but its median wait among transplanted patients is among the shortest.

Interpretation

Region should be included in modeling and fairness evaluation, but raw regional differences are not automatically causal. Donor availability, center practices, patient mix, blood type, listing year, and follow-up may all contribute.

13. Listing-Center Variation

The dataset includes 256 listing centers. To reduce instability from tiny centers, center-level comparisons were restricted to the 182 centers with at least 500 registrations.

Metric across centers >=500 registrations	Minimum	Median	Maximum
Adverse-event rate	2.72%	14.02%	34.55%
Transplant rate	16.04%	45.75%	81.60%
Still-waiting rate	0.00%	21.20%	54.21%

These raw differences are very large. For example, the highest adverse rate is more than ten times the lowest. However, centers serve different patient populations and operate in different regional contexts.

- Candidate age and severity
- Functional status and dialysis exposure
- Blood type and sensitization
- Race/ethnicity composition
- Donor availability and region
- Listing year and follow-up duration
- Center-specific practices and referral patterns

Important caution

These center comparisons show heterogeneity worth investigating; they do not prove that a particular center performs better or worse. Risk-adjusted analysis is required before making center-performance claims.

14. Transplant-Time Fields, Consistency, and Leakage

Several fields are populated only at or after transplantation or near the end of the waitlist episode. These can support descriptive EDA but are not valid listing-time predictors.

- End CPRA ('END_CPRA')
- End Medical Status ('END_STAT')
- Transplant Date ('TX_DATE')
- Donor Type ('DON_TY')
- Transplanted Organ ('ORGAN')

- Transplant Event ID ('TRR_ID_CODE')
- Donor ID ('DONOR_ID')
- Patient Survival Time ('PTIME')
- Patient Status ('PSTATUS')
- Composite Death Date ('COMPOSITE_DEATH_DATE')

Transplant-time field consistency

Transplant Date ('TX_DATE') is present for 230,749 registrations labeled transplanted but missing for 3,680 additional transplanted registrations. Six registrations labeled died also have a transplant date. Donor Type ('DON_TY') follows the same pattern. Therefore, transplant-date presence alone should not define the transplant outcome; the derived event/outcome fields should remain primary.

Composite Death Date is not waitlist death

Composite Death Date ('COMPOSITE_DEATH_DATE') is present for every registration whose waitlist outcome is died, but it is also present for other outcomes, including 27,818 transplanted registrations. A patient can receive a transplant and die later. Therefore, waitlist outcome = died and ever having a death date are different concepts.

Donor Type ('DON_TY')

Among registrations classified as transplanted, donor type counts are: C = 167,046; L = 63,702; missing = 3,680; F = 1. Any unusual code such as F should be checked against the official OPTN dictionary before interpretation. Donor Type is known at transplant and is not a valid baseline predictor.

Multi-organ and diagnosis fields

Multi-organ Listing Flag ('MULTIORG') is missing in 483,004 registrations and Y in 11,858. Its sparsity likely reflects conditional applicability rather than general data failure. Kidney Diagnosis ('DIAG_KI') contains roughly 70 observed values and is missing in more than half the dataset; because the project dictionary treats it as a later field in this extract, it should not enter the listing-time model unless a separate baseline diagnosis variable is verified.

15. Modeling Readiness and Leakage-Safe Feature Set

Recommended listing-time feature set

- Age at Listing ('INIT_AGE')
- Sex ('GENDER')
- Race/Ethnicity ('ETHCAT')
- Blood Type ('ABO')
- Initial CPRA ('INIT_CPRA')
- On Dialysis at Listing ('ON_DIALYSIS')
- Body Mass Index ('BMI_TCR')
- Functional Status ('FUNC_STAT_TCR')
- Initial Medical Status ('INIT_STAT')
- OPTN Region ('REGION')

A derived Listing Year or policy-era feature may also be useful because of strong temporal shifts. Pre-listing dialysis duration may contain valid prior information, but it should be discussed with the advisor before being added to the official modeling feature set.

Variables that should not enter the baseline classification model

- Removal Reason ('REM_CD')

- Outcome (`outcome`)
- Adverse Event (`event\_adverse`) except as the target
- Transplant Event (`event\_transplant`)
- Censored (`censored`)
- Derived Time to Outcome (`days\_to\_event`)
- Total Days on Waitlist (`DAYSWAIT\_CHRON`)
- End CPRA (`END\_CPRA`)
- End Medical Status (`END\_STAT`)
- Transplant Date (`TX\_DATE`)
- Donor Type (`DON\_TY`)
- Transplanted Organ (`ORGAN`)
- Patient Survival Time (`PTIME`)
- Patient Status (`PSTATUS`)
- Composite Death Date (`COMPOSITE\_DEATH\_DATE`)
- Donor ID (`DONOR\_ID`)
- Transplant Event ID (`TRR\_ID\_CODE`)

16. Data-Cleaning Plan Before Modeling

1. Preserve the raw analytic dataframe unchanged and create a separate cleaned/modeling copy.
2. Convert listing, end, dialysis, transplant, and composite death date fields to proper datetime types.
3. Keep Age at Listing unchanged; ages 0-17 are legitimate pediatric registrations.
4. Investigate Body Mass Index with domain guidance and convert clearly impossible values to missing rather than allowing them to distort the model.
5. Investigate the single Functional Status = 1 record and likely treat it as missing if not supported by the source coding scheme.
6. Exclude or correct the 59 registrations with End Date before Listing Date for survival modeling.
7. Do not blindly median-impute Initial CPRA; its missingness is strongly time-dependent.
8. Consider a missingness indicator for Initial CPRA and compare models with and without CPRA.
9. Encode categorical codes such as Race/Ethnicity, Region, Initial Medical Status, and Functional Status appropriately instead of treating raw numeric codes as ordinary continuous values.
10. Keep patient and registration identifiers out of the feature matrix.
11. Use Patient ID (`PT\_CODE`) to prevent the same person from appearing in both training and evaluation sets.

17. Validation, Fairness, and Evaluation Strategy

A simple random row-level split is not recommended. The evaluation design should account for both repeated patients and time.

- Group by Patient ID (`PT\_CODE`) so the same patient does not appear across train and test.
- Evaluate performance on newer listing years separately.
- Compare model behavior before and after the major Initial CPRA missingness shift beginning in 2023.
- Consider temporal validation to test whether models trained on earlier years generalize to later candidates.
- Report subgroup performance by age, sex, race/ethnicity, blood type, and region.

Fairness considerations

Raw subgroup outcome rates differ across demographic and geographic groups, but these differences should not automatically be labeled bias. A proper fairness analysis should determine whether the model performs

equally well, is similarly calibrated, systematically overpredicts or underpredicts risk, or has materially different sensitivity and false-negative rates across groups.

18. Key Risks Identified During EDA

Risk	Why it matters
Censoring	More than one-fifth of registrations are still waiting; censoring is extremely high in recent years.
Temporal bias	Recent cohorts have shorter follow-up, making raw event rates look artificially lower.
Patient leakage	The same patient can have multiple registrations.
Target leakage	Many fields are only known after transplant or at the end of the episode.
CPRA missingness shift	Initial CPRA becomes approximately 80% missing beginning in 2023.
BMI errors	A small set of impossible values can distort means and model fitting.
Geographic/center heterogeneity	Large differences require careful adjustment before interpretation.
Class imbalance	Only 14.09% of registrations are adverse positives.
Documentation drift	The current CSV contains 38 columns while some project documents describe more.

19. Main Story of the Dataset

The dataset represents nearly half a million kidney waitlist registrations from 2015 through mid-2026. Approximately 47% resulted in a transplant through that registration, roughly 14% ended in death or removal because the patient became too sick, and about 21% are still waiting and therefore censored.

The patients at highest observed risk tend to be older, already on dialysis, and in poorer functional condition at listing. Geography also matters: OPTN region and individual listing center show substantial variation in raw outcomes. Blood type is especially relevant to the transplant process, with O and B candidates showing lower observed transplant rates and longer waits than A and AB candidates.

However, this dataset cannot be treated like an ordinary independent classification table. The same patient can appear more than once, many recent patients have incomplete follow-up, several fields reveal future information, and Initial CPRA undergoes a major missingness shift after 2022. The central modeling challenge is therefore not simply finding an algorithm with a high AUC; it is building a model that uses only information available at listing, correctly handles censoring and repeated patients, remains stable across time, and performs fairly across demographic and geographic groups.

20. Recommended Next Steps

1. Freeze and document the raw analytic dataset.
2. Create a cleaned modeling copy rather than modifying the raw dataframe.
3. Resolve or flag invalid BMI and date values.
4. Convert variable types correctly and document every transformation.
5. Finalize the listing-time feature set.
6. Decide how Initial CPRA missingness will be handled.
7. Define a patient-grouped and preferably time-aware train/validation/test strategy.
8. Establish an interpretable logistic-regression baseline for adverse-event classification.
9. Carefully define how censored registrations will be handled in the classification task.

10. Build the survival-analysis dataset using Transplant Event ('event_transplant') and Derived Time to Outcome ('days_to_event').
11. Compare Cox proportional hazards with a competing-risks approach where appropriate.
12. Evaluate subgroup discrimination, calibration, and error rates.
13. Investigate geographic and center-level differences using adjusted rather than raw comparisons.
14. Document every preprocessing, exclusion, and evaluation decision for reproducibility.

21. Final EDA Conclusion

The OPTN kidney waitlist analytic dataset is large, rich, and appropriate for the project's classification, survival, explainability, and fairness objectives. The EDA identified strong predictive relationships with age, dialysis status, functional status, blood type, and geography, while also uncovering important structural challenges including censoring, repeated patients, temporal shifts in data availability, extreme BMI values, invalid date records, and substantial center-level heterogeneity.

The most important lesson is that time and data availability must be treated as part of the problem itself. A model that ignores censoring, future-information leakage, repeated registrations, or the CPRA recording shift could produce apparently strong metrics while giving misleading conclusions.

Recommended transition

The project is ready to move from EDA into a carefully documented preprocessing stage, followed by leakage-safe baseline classification and survival models. More complex models should come only after these baselines and validation rules are stable.

Appendix A. Compact EDA Reference

Finding	Value / interpretation
Rows	494,862 waitlist registrations
Unique patients	414,187
Primary adverse target	69,743 (14.09%)
Transplanted	234,429 (47.37%)
Still waiting / censored	103,300 (20.87%)
All 10 listing features present	350,337 (70.79%)
Initial CPRA missing	28.84% overall; ~80% in 2023-2026
BMI issues	240 ≤10; 474 >60; 263 >100; 124 >1000
Invalid end-before-start dates	59
On-dialysis adverse rate	15.51% vs 10.28% off dialysis
Age pattern	Adverse rate increases strongly with age
Functional status pattern	Better function generally corresponds to lower adverse rate
Blood type pattern	O/B lower transplant rates and longer waits than A/AB
Regional variation	~5.4 pp adverse-rate spread; >13 pp transplant-rate spread
Center variation (≥500 cases)	Adverse 2.72%-34.55%; transplant 16.04%-81.60%
Critical split rule	Group by Patient ID ('PT_CODE')
Critical modeling concern	Censoring and post-2022 CPRA missingness